

Revised Problem Statement

Multi-Task Analysis of Telugu-English Code-Mixed News Comments

1. Background and Motivation

The rapid growth of Telugu-language news content on digital platforms such as YouTube has resulted in large volumes of user-generated comments that frequently exhibit Telugu-English code-mixing (commonly referred to as *Tenglish*). In these comments, users alternate between Telugu script, Romanized Telugu, and English within a single utterance, creating a highly informal and linguistically complex form of communication.

Examples include comments such as:

- "CM garu cheppindi correct kadhu, complete fake agenda"
- "Bro idhi chala dangerous situation, meeru artham cheskovadam ledu"

Such code-mixed content presents significant challenges for Natural Language Processing (NLP) systems because standard models are generally trained on monolingual Telugu or English corpora and struggle to handle mixed scripts, transliteration variations, informal spelling, and social-media-specific language patterns.

Despite the increasing prevalence of Telugu-English code-mixed content and its relevance to public discourse, online toxicity, and political communication, Telugu remains comparatively underrepresented in code-mixed NLP research. Existing benchmark datasets primarily focus on Tamil-English and Kannada-English code-mixed text, while Telugu-English resources remain limited.

2. Research Gap

This study is motivated by the following gaps in existing literature:

1. To the best of our knowledge, no publicly available Telugu-English code-mixed YouTube news comment dataset jointly annotated for sentiment, toxicity, and stance currently exists.
2. Existing Telugu and code-mixed NLP studies typically treat sentiment analysis, toxicity detection, and stance detection as independent tasks rather than exploring a unified multi-task framework.
3. Limited comparative evaluation exists for multilingual transformer models such as mBERT, XLM-R, IndicBERT, and MuRIL on Telugu-English code-mixed news comments.
4. Explainability analysis of multilingual transformer predictions on Telugu-English code-mixed text remains largely unexplored.

3. Research Objectives

The proposed study aims to address the following research questions:

RQ1: How effectively can multilingual transformer models (mBERT, XLM-R, IndicBERT, and MuRIL) perform sentiment detection, toxicity detection, and stance detection on Telugu-English code-mixed news comments?

RQ2: To what extent does a multi-task learning framework, where all tasks share a common transformer encoder, outperform independently trained single-task models?

RQ3: Which Telugu and English lexical features contribute most strongly to model predictions, as revealed through explainability techniques such as SHAP and attention analysis?

RQ4: What impact does transliteration-aware normalization have on downstream classification performance?

4. Proposed Tasks

The study will investigate three complementary NLP tasks:

Sentiment Detection

Classification of comments into:

- Positive
- Negative
- Neutral

Toxicity Detection

Classification of comments into:

- Toxic
- Not Toxic
- Ambiguous

Stance Detection

Classification of comments into:

- Support
- Oppose
- Neutral

Stance labels will be assigned relative to the news topic represented by the video title and associated context.

5. Annotation Protocol

- Annotation Unit: Individual YouTube comments (top-level comments and replies)
- Context Provided: Video title and comment text

- Labels per Comment: Sentiment, Toxicity, and Stance
- Primary Annotator: Aladurthi Sai Meghanath
- Secondary Annotator: To be identified
- Inter-Annotator Agreement: Cohen's κ on a 200-comment subset
- Target Agreement: $\kappa \geq 0.70$

A browser-based annotation tool has been developed to support multi-task annotation, progress tracking, and stance annotation with video-title context.

6. Dataset Collection Plan

A YouTube Data API-based collection pipeline has been developed to gather comments from major Telugu news channels.

Current corpus size:

- 2,316 comments

Planned expansion:

- 8,000–10,000 comments

The expanded dataset will include comments from diverse news categories, including:

- Politics
- Social Issues
- Crime
- Entertainment
- Sports

7. Expected Contributions

The proposed research is expected to contribute:

1. A Telugu-English code-mixed YouTube news comment dataset annotated for sentiment, toxicity, and stance.
2. A comparative evaluation of multilingual transformer models for low-resource code-mixed NLP.
3. A multi-task learning framework for jointly modeling sentiment, toxicity, and stance.
4. An explainability analysis of multilingual transformer behavior on Telugu-English code-mixed text.

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